

| Question | Answer                                                                                                                                                                                                                                                                                                                                  | Marks                         |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| 6(a)     | <p>Frequency = 50 Hz</p> <p>Peak Voltage = 340 V</p> <p>Root-mean-square voltage = 240 V</p>                                                                                                                                                                                                                                            | <p>A1</p> <p>A1</p> <p>A1</p> |
| 6(b)(i)  | Voltage = $240/20 = 12 \text{ V}$                                                                                                                                                                                                                                                                                                       | A1                            |
| 6(b)(i)  | <p>Power consumed by 6.0 resistor, <math>P = \frac{V^2}{R} = \frac{12.0^2}{6.0} = 24.0 \text{ W}</math></p> <p>Power generated by primary circuit, <math>P = \frac{24.0}{0.95} = 25.263 \text{ W}</math></p> <p>Current in the primary circuit, <math>I = \frac{P}{V} = \frac{25.263}{240} = 0.10526 \approx 0.105 \text{ A}</math></p> | <p>A1</p> <p>A1</p> <p>A1</p> |

|  |           |   |
|--|-----------|---|
|  | Max Marks | 7 |
|--|-----------|---|

| Setter: Soo Yen<br>Marker: |                      |       |
|----------------------------|----------------------|-------|
| Question                   | Answer               | Marks |
| 7(a)                       | Photoelectric Effect | A1    |